

(TR-104) PYTHON WITH AI/ML

Training Day 86 Report

Date: 11 May 2026

Topic Covered: Backend Optimization, AI Response Enhancement, Streamlit Development & Supabase Integration

The eighty-sixth day of training focused on optimizing backend performance, improving AI response generation, enhancing system maintainability, initiating college project development using Streamlit, and exploring Supabase database integration.

- Implemented dynamic tool loading mechanisms to optimize agent performance and reduce unnecessary processing.
- Reduced prompt size and improved backend efficiency through selective tool binding strategies.
- Added response truncation handling and optimized counting operations for large datasets.
- Resolved rate-limit and token-size issues through backend performance optimization techniques.
- Enhanced tool logic by improving parameter handling and MongoDB query processing.
- Updated backend tools using LLM-friendly naming conventions to improve AI tool selection accuracy.
- Improved database query reliability and response accuracy across multiple backend services.
- Enhanced AI response formatting, grammar handling, and conversational output quality.
- Improved frontend readability by optimizing rendering behavior for count-based and analytical responses.
- Refactored backend modules to improve code readability, organization, and maintainability.
- Enhanced intent categorization and routing logic for better query handling.
- Initiated the college project setup by creating the project structure and developing the Streamlit home screen.

- Connected multiple frontend screens and planned backend function implementation for future development.
- Created project-related tables in Supabase and analyzed database interaction workflows.
- Reviewed Supabase query handling, database structure, and backend integration planning.
- Explored image processing libraries and studied the functionalities of the dlib library.
- Researched bcrypt for password hashing and secure authentication implementation.
- Revised concepts related to token optimization, rate-limit handling, dynamic tool loading, Streamlit development, Supabase integration, image processing, and password security mechanisms.

Overall, the training enhanced understanding of backend optimization, AI response management, scalable application architecture, Streamlit-based frontend development, database integration using Supabase, and security-focused development practices.